Supplementary Information

Ontology-constrained multi-LLM scoring of hypothesis support in the predictive processing literature

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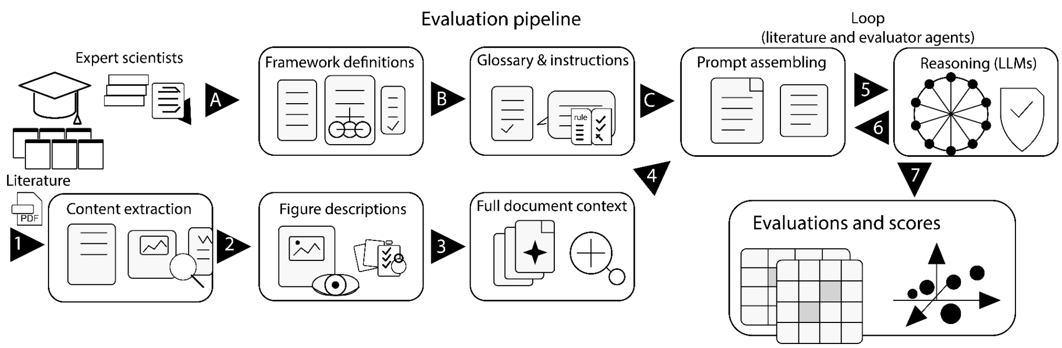
Supplementary Information

Supplementary Figures

Supplementary Fig. S1

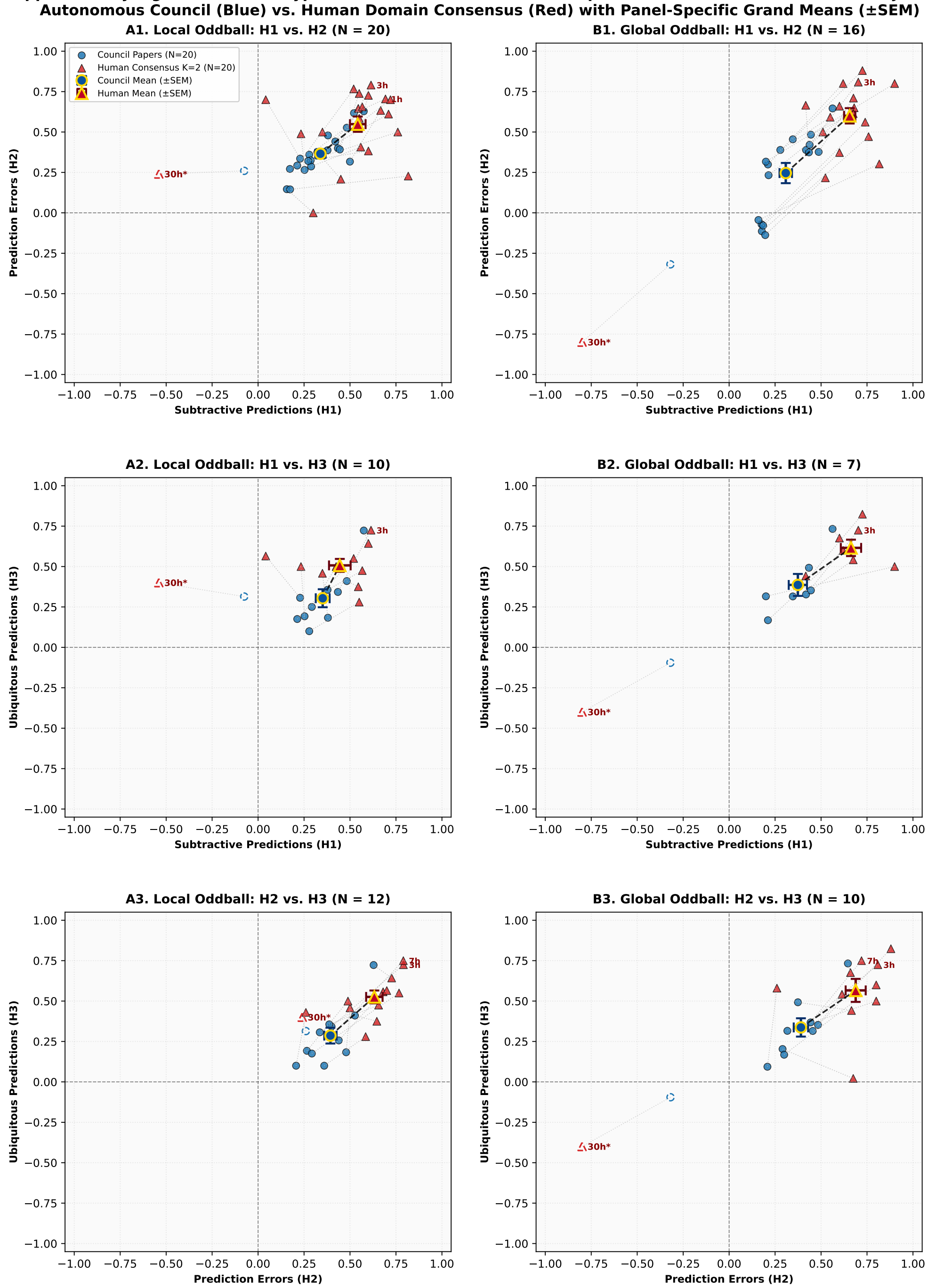
Supplementary Fig. S1: Evaluation Pipeline Architecture. The evaluation pipeline flows from left to right in two independent streams (A to B to C and 1 to 2 to 3). The A to B to C pathway establishes the instructions, canonical glossary, and prompt for the models. The 1 to 2 to 3 pathway ingests the paper and combines figures with text to provide a single text object for the models. Steps 5 and 6 loop between models and papers until all evaluations are completed. The final step 7 quantifies and visualizes the evaluations.



Supplementary Fig. S2

Supplementary Fig. S2: ~~**Independent Human Expert Concordance (**$K = 2$ **domain experts) . (A)Factor-cell level concordance between human consensus (**$N = 175$ **non-null paired factor cells)**~~ **Pairwise 2D Hypothesis-Space Projections and** ~~**the autonomous LLM council consensus (**$r = 0.162, \rho = 0.241$**, MAE** $= 0.301$**, Directional Concordanc . (B)Study-level aggregation across** $N = 25$ **common evaluable papers(Pearson** $r = 0.529$**, Spearman** $\rho = 0.492$**) , with stronger study-level than factor-cell concordance**~~**Human– Concordance**. Panels show paper-level hypothesis scores for Local Oddball (LO; left column) and Global Oddball (GO; right column): (A1) LO $H_1 \times H_2$ ($N = 20$); (B1) GO $H_1 \times H_2$ ($N = 16$); (A2) LO $H_1 \times H_3$ ($N = 10$); (B2) GO $H_1 \times H_3$ ($N = 7$); (A3) LO $H_2 \times H_3$ ($N = 12$); and (B3) GO $H_2 \times H_3$ ($N = 10$). Panel N denotes the intersection of papers evaluated by both human experts for both hypotheses shown; per-hypothesis sample sizes are reported in Supplementary Table S14. Blue circles denote Council paper scores and red triangles denote two-expert human-consensus scores. Large gold-bordered markers show panel-specific means ± SEM across the same independent papers. Dotted gray lines connect paired evaluations for the same study. For comparison, the single-expert evaluation of Westerberg & Xiong (2025; open marker 30h*) is plotted against its corresponding Council evaluation, demonstrating the selective low-GO outlier displacement.

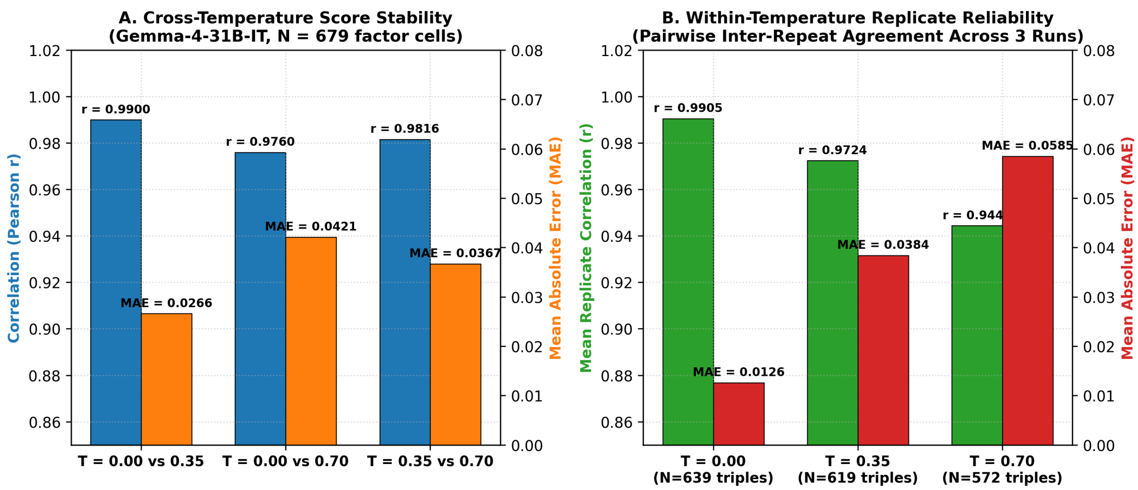
Supplementary Fig. S2: Pairwise Hypothesis Concordance across Independent Literature Studies (3×2 Projection)
Autonomous Council (Blue) vs. Human Domain Consensus (Red) with Panel-Specific Grand Means (±SEM)



Panels display exact K=2 intersection papers evaluated by both domain experts for the two plotted hypotheses (A1: N=20, B1: N=16, A2: N=10, B2: N=7, A3: N=12, B3: N=10).
Large gold-bordered markers show panel-specific Human and Council Grand Means ± across-study SEM (SEM = SD_papers / sqrt(N_panel)). Dotted grey lines connect paired evaluations.
30h*: Westerberg & Xiong (2025) single-expert benchmark (open marker; LO: [-0.54, +0.50, +0.40], GO: [-0.80, -0.80, -0.40]). Landmark K=2 studies: 01h: Attinger2017 | 03h: Bastos2012 | 07h: Friston2010.

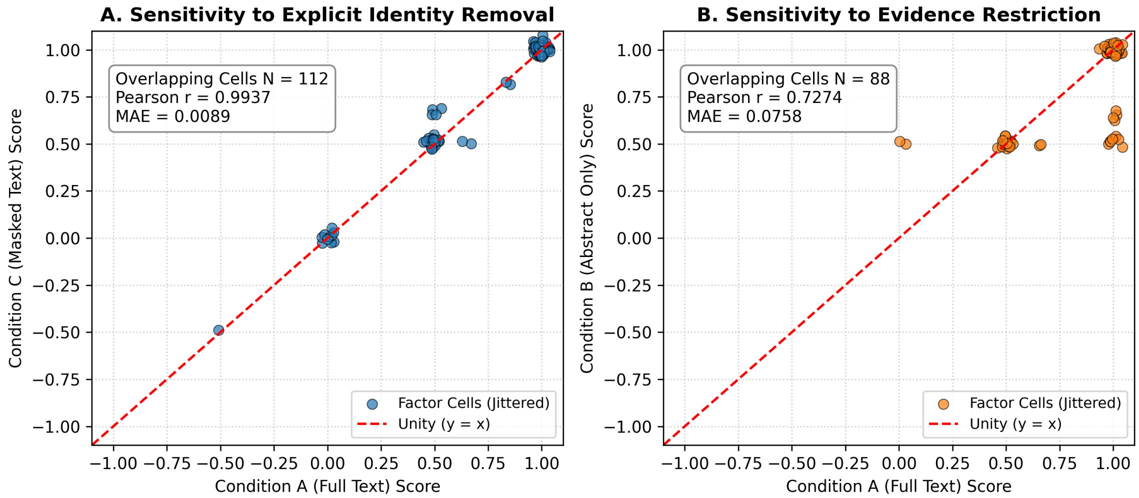
Supplementary Fig. S3

Supplementary Fig. S3: ~~837-Call Factorial Robustness Analysis (31 papers × 3 models × 3 temperatures × 3 repeats)~~Decoding Robustness and Stochastic Replicate Reliability Analysis. (A) ~~Score distribution stability~~ Cross-temperature score stability in the primary complete evaluation model (`gemma-4-31b-it`; 279 calls across 31 papers), showing Pearson correlations ($r \geq 0.9760$) and mean absolute errors (MAE ≤ 0.0421) across sampling temperatures ~~(~~$T \in \{0.00, 0.35, 0.70\}$~~) for OLMo-3-32B-Think, Gemma-4-31B-IT, and Phi-4-Reasoning-Plus~~. (B) ~~Context displacement stability (~~$\Delta_{\text{LO-GO}}$~~)demonstrating that the primary scientific contrast remains stable across the tested decoding settings and stochastic repeats~~Replicate reliability across independent stochastic generations held at fixed temperatures ($T = 0.00 : \bar{r} = 0.9905, \text{MAE} = 0.0126$; $T = 0.35 : \bar{r} = 0.9724, \text{MAE} = 0.0384$; $T = 0.70 : \bar{r} = 0.9444, \text{MAE} = 0.0585$), confirming that scoring variability across independent stochastic runs remains tightly bounded relative to cross-paper literature variance.



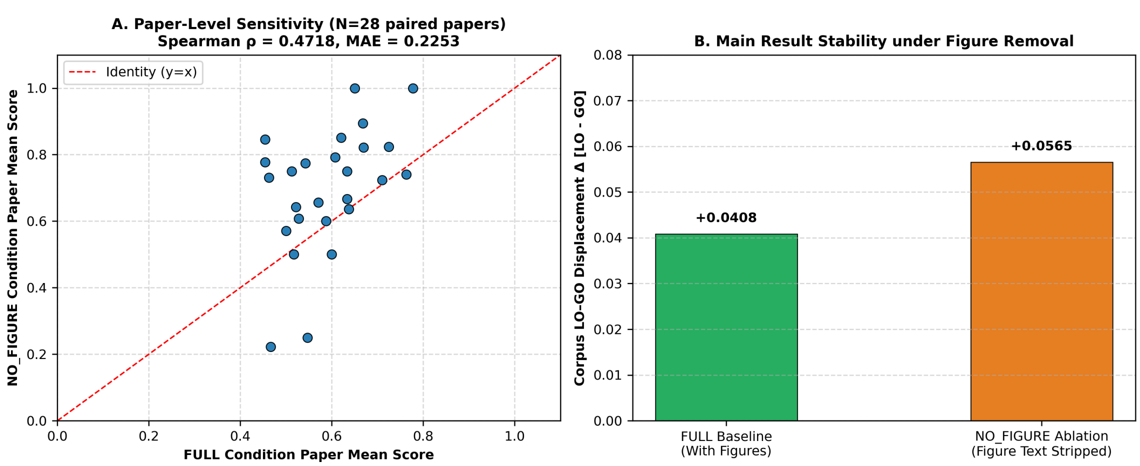
Supplementary Fig. S4

Supplementary Fig. S4: Evidence Grounding ~~vs. Prior Bias Sensitivity Analysis(~~Control and Shuffled Text Sensitivity Analysis. Evaluates 54 inference calls across intact manuscript text (FULL), ~~abstract-only, and masked-manuscript evidence controls). (A)Sensitivity to explicit study/author identity removal (Condition A Full Text vs Condition C Masked Text: Pearson~~ ~~$r = 0.9937$~~~~, MAE =~~ ~~0.0089~~ ~~across~~ ~~$N = 112$~~ ~~overlapping factor cells), showing minimal sensitivity to explicit study/author identity removal. (B)Sensitivity to evidence restriction (Condition A Full Text vs Condition B Abstract Only: Pearson~~ ~~$r = 0.7274$~~~~, MAE =~~ ~~0.0758~~ ~~across~~ ~~$N = 88$~~ ~~overlapping factor cells~~shuffled-evidence text (H_0), and prior-only prompts without paper text (PRIOR). Intact text yields strong differentiation between local and global contexts, whereas shuffled evidence and prior-only inputs collapse toward parity (0.450–0.493), showing ~~score changes when detailed empirical text is withheld. Full, Shuffled, and Prior-only condition summaries are reported in Table S10.~~ sensitivity to supplied empirical evidence beyond prior-only input.



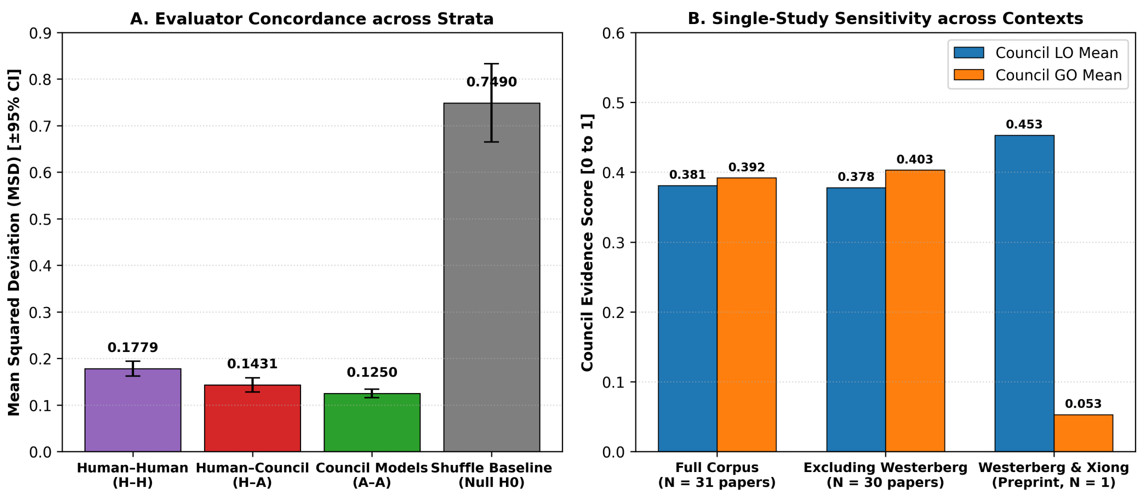
Supplementary Fig. S5

Supplementary Fig. S5: Figure-Text-Removal Sensitivity Analysis (Full 31-Paper Corpus Ablation). (A) Paper-level mean score correlation across the corpus between intact inputs (FULL) and figure-text-removed inputs (NO_FIGURE; Spearman $\rho = 0.4718$ ~~,~~ across $N = 28$ paired empirical paper means; factor-cell level MAE = 0.2253 across all 533 paired ~~factor~~ cells nested in 31 papers). (B) ~~Corpus-level LO − GO context displacement stability across 28 paired empirical papers ($\Delta_{\mathrm{Full}} = +0.0408$ vs $\Delta_{\mathrm{NoFig}} = +0.0565$;~~ Factor-cell level directional concordance and stability across the corpus (directional concordance = 99.42% on non-zero cells [518/521], all-cell sign agreement = 97.19% [518/533]~~), showing that the corpus-level contrast retained its direction despite~~, Pearson $r = 0.8539$), demonstrating that while removing figure descriptions produces measurable paper-level score ~~changes~~adjustments, overall factor directionality was preserved across the literature corpus, while paper-level rank correspondence was moderate.



Supplementary Fig. S6

Supplementary Fig. S6: Evaluator ~~Strata Mean Squared Deviation~~ Concordance Strata and Single-Study ~~Westerberg Sensitivity~~ Sensitivity Analysis. (A) ~~Evaluator strata mean squared deviation (MSD ± 95% CI) across human–human inter-rater reliability (H–$H = 0.1779$), human consensus vs. autonomous~~ Mean squared difference (MSD) across evaluator strata ($K = 2$ human domain experts and autonomous LLM council) compared against a null permutation reference distribution (H_0, $B = 10{,}000$ iterations). Agreement within human experts (H–$H = 0.1779$) and between human consensus and council (H–A = 0.1431) ~~, council inter-model agreement (A–$A = 0.1250$), and the permutation shuffle baseline (Null $H_0 = 0.7490$~~is significantly tighter than chance ($H_0 = 0.7490$, $p_{\text{perm}} = 0.0001$, 0/10,000 extreme permutations). (B) ~~Single-study sensitivity of corpus displacement ($\Delta_{\text{LO-GO}}$ ± 95% CI) across the full paired corpus ($N = 28$, $\Delta = -0.0177$~~Council condition means for Local Oddball (LO) and Global Oddball (GO) across the entire corpus ($N = 31$), excluding Westerberg & Xiong (2025~~)~~ ~~($N = 27$, $\Delta = +0.0297$)~~; $N = 30$), and for Westerberg & Xiong alone (~~2025) alone ($N = 1$, $\Delta = -1.2985$)~~N = demonstrating that the aggregate corpus contrast is robust across published studies and that the preprint represents a selective low-GO outlier.



Supplementary Tables

~~**Factor Number Factor Name Definition Context Scope** 1 Subtractive Inhibition (SST) Linear input suppression via somatic inhibition (Somatostatin-mediated). LO+GO 2 Divisive Inhibition (PV) Multiplicative gain reduction (Parvalbumin-mediated). LO+GO 3 Inhibition (GABA) Chloride-mediated inhibition for prediction suppression. LO+GO 4 Habituation to Sequence Habituation suppresses local oddball response relative to a novel local oddball. LO 5 Synaptic Depression (Adaptation) Passive fatigue of synaptic efficacy due to repetition of stimulus. LO 6 Activity Suppression Reduction in firing rates for expected (predictable) stimuli. LO+GO 7 Selective Sharpening Signal-to-noise increase for predictable stimuli, where noise is selectively suppressed to highlight relevant information. LO+GO 8 Alpha/Beta Mediated Suppression Low-frequency bands (8-30Hz) associated with predictions, inhibiting prediction error signals. LO+GO 9 VIP-Mediated Disinhibition VIP-mediated inhibition of SST/PV neurons, leading to disinhibited pyramidal activity during prediction error. LO+GO 10 Precision Weighting (Gain) Top-down amplification of error units, leading to disinhibited gain in response to attended or highly precise stimuli. LO+GO 11 E/I Balance Shift Dynamic adjustment toward higher inhibition for predictable stimuli, leading to suppressed neural firing. LO+GO 12 Omission Response A generated internal signal due to absence of expected input LO+GO~~

#	Factor Name	Definition
1	Subtractive Inhibition (SST)	Linear input suppressi (Somatostatin-mediate
2	Divisive Inhibition (PV)	Multiplicative gain rec
3	Inhibition (GABA)	Chloride-mediated inh
4	Habituation to Sequence	Habituation suppresse to a novel local oddba
5	Synaptic Depression (Adaptation)	Passive fatigue of syna stimulus.
6	Activity Suppression	Reduction in firing rat stimuli.
7	Selective Sharpening	Signal-to-noise increas noise is selectively sup information.
8	Alpha/Beta Mediated Suppression	Low-frequency bands predictions, inhibiting
9	VIP-Mediated Disinhibition	VIP-mediated inhibitio disinhibited pyramidal
10	Precision Weighting (Gain)	Top-down amplificatio disinhibited gain in re precise stimuli.
11	E/I Balance Shift	Dynamic adjustment t predictable stimuli, lea
12	Omission Response	A generated internal s input.

Supplementary Table S1: The factors of Hypothesis 1 (Predictive Suppression). LO =

local oddball. GO = global oddball.

~~**Factor Number Factor Name Definition Context Scope** 13 Error Unit Depolarization Supragranular pyramidal excitation driving ascending error signals. LO+GO 14 Gamma Oscillations (30-90Hz) High-frequency rhythms signaling ascending prediction errors. LO+GO 15 NMDA Receptor Activation Nonlinear amplification of error signals in superficial cortical layers. LO+GO 16 Feedforward Laminar Flow Supragranular-to-granular/supragranular ascending anatomical projection of errors. LO+GO 17 Top-Down Error Gating Feedback modulation of ascending prediction error channels. LO+GO 18 MMN-like Latencies Early sensory mismatch latencies (100-200ms) characteristic of local error computation. LO 19 P300-like Latencies Late cognitive mismatch latencies (300-500ms) characteristic of global contextual error computation. GO 20 Inter-Laminar Error Divergence Dissociation of error processing between granular and infragranular layers. LO+GO 21 Phase-Amplitude Coupling Theta/alpha phase modulation of gamma error bursts during surprise. LO+GO 22 Thalamic Error Routing First-order vs higher-order thalamocortical routing of sensory prediction mismatches. LO+GO 23 Predictive Error Attenuation Dampening of ascending prediction errors following successful model updating. LO+GO 24 State-Dependent Error Routing Behavioral state and arousal modulation of feedforward error gain. LO+GO~~

#	**Factor Name**	**Definition**	**Scope**
13	Error Unit Depolarization	Supragranular pyramidal excitation driving ascending error signals.	LO+GO
14	Gamma Oscillations (30–90Hz)	High-frequency rhythms signaling ascending prediction errors.	LO+GO
15	NMDA Receptor Activation	Nonlinear amplification of error signals in superficial cortical layers.	LO+GO
16	Feedforward Laminar Flow	Supragranular-to-granular/supragranular ascending anatomical projection of errors.	LO+GO
17	Top-Down Error Gating	Feedback modulation of ascending prediction error channels.	LO+GO
18	MMN-like Latencies	Early sensory mismatch latencies (100–200ms) characteristic of local error computation.	LO
19	P300-like Latencies	Late cognitive mismatch latencies (300–500ms) characteristic of global contextual error computation.	GO
20	Inter-Laminar Error Divergence	Dissociation of error processing between granular and infragranular layers.	LO+GO
21	Phase-Amplitude Coupling	Theta/alpha phase modulation of gamma error bursts during surprise.	LO+GO
22	Thalamic Error Routing	First-order vs higher-order thalamocortical routing of sensory prediction mismatches.	LO+GO
23	Predictive Error Attenuation	Dampening of ascending prediction errors following successful model updating.	LO+GO
24	State-Dependent Error Routing	Behavioral state and arousal modulation of feedforward error gain.	LO+GO

Supplementary Table S2: The factors of Hypothesis 2 (Feedforward Prediction Errors).

~~**Factor Number Factor Name Definition Context Scope** 25 Cross-Modal Ubiquity Implementation of predictive coding mechanisms across multiple sensory modalities (visual, auditory, somatosensory). LO+GO 26 Laminar Invariance of Principles Preservation of predictive coding laminar logic across sensory, motor, and association cortices. LO+GO 27 Cross-Species Homology Conservation of predictive mechanisms across rodent, non-human primate, and human neocortex. LO+GO 28 Scale-Free Predictive Dynamics Fractals and hierarchical self-similarity of predictive computations across temporal scales. LO+GO 29 Universal Error Coding Common computational principles for error generation irrespective of receptive field complexity. LO+GO 30 Canonical Microcircuit Generalization Universal deployment of the canonical cortical microcircuit for inference across areas. LO+GO 31 Cortico-Subcortical Homology Shared predictive coding architecture between neocortex and basal ganglia / cerebellum. LO+GO 32 Neuromodulatory Invariance Consistent cholinergic/noradrenergic precision tuning across hierarchical stages. LO+GO 33 Temporal Stability of Ubiquity The consistent and non-transient presence of these mechanisms across the hierarchy over long recording sessions. LO+GO 34 Hierarchical Order Stability Evidence that the predictive mechanism is ubiquitous and not localized to a single hierarchical pole. LO+GO 35 Population-Wide Ubiquity Consistency of the mechanism across diverse neural populations and cell types throughout the hierarchy. LO+GO 36 State-Independent Ubiquity Presence of the mechanism across different brain states (e.g., wakefulness, sleep, anesthesia) throughout the hierarchy. LO+GO~~

#	Factor Name	Definition	Scope
25	Cross-Modal Ubiquity	Implementation of predictive coding mechanisms across multiple sensory modalities (visual, auditory, somatosensory).	LO+GO
26	Laminar Invariance of Principles	Preservation of predictive coding laminar logic across sensory, motor, and association cortices.	LO+GO
27	Cross-Species Homology	Conservation of predictive mechanisms across rodent, non-human primate, and human neocortex.	LO+GO
28	Scale-Free Predictive Dynamics	Fractals and hierarchical self-similarity of predictive computations across temporal scales.	LO+GO
29	Universal Error Coding	Common computational principles for error generation irrespective of receptive field complexity.	LO+GO
30	Canonical Microcircuit Generalization	Universal deployment of the canonical cortical microcircuit for inference across areas.	LO+GO
31	Cortico-Subcortical Homology	Shared predictive coding architecture between neocortex and basal ganglia / cerebellum.	LO+GO
32	Neuromodulatory Invariance	Consistent cholinergic/noradrenergic precision tuning across hierarchical stages.	LO+GO
33	Temporal Stability of Ubiquity	The consistent and non-transient presence of these mechanisms across the hierarchy over long recording sessions.	LO+GO
34	Hierarchical Order Stability	Evidence that the predictive mechanism is ubiquitous and not localized to a single hierarchical pole.	LO+GO
35	Population-Wide Ubiquity	Consistency of the mechanism across diverse neural populations and cell types throughout the hierarchy.	LO+GO
36	State-Independent Ubiquity	Presence of the mechanism across different brain states (e.g., wakefulness, sleep, anesthesia) throughout the hierarchy.	LO+GO

Supplementary Table S3: The factors of Hypothesis 3 (Ubiquity).

~~**Term/Context Description Definition (notation)** Oddball A surprising event, often a mismatch = {B}$given = \{A\}$ Local Oddball (LO) An oddball that is immediate or short-term, often confounded by low-level adaptation mechanisms. = {AB}$given = \{AA\}$ Global Oddball (GO) An oddball that is not LO, often contextual, that is not caused by adaptation mechanisms. = {AA}$given = \{AB\}$~~

Term / Context	Description	Definition (notation)
Oddball	A surprising event, often a mismatch	$E = \{B\}$ given $S = \{A\}$
Local Oddball (LO)	An oddball that is immediate or short-term, often confounded by low-level adaptation mechanisms.	$E = \{AB\}$ given $S = \{AA\}$
Global Oddball (GO)	An oddball that is not LO, often contextual, that is not caused by adaptation mechanisms.	$E = \{AA\}$ given $S = \{AB\}$

Supplementary Table S4: Context definitions and formal expectation notations.

~~**# Study Identifier Modality / Preparation Inclusion Basis Bibliographic Citation** 1 Attinger et al., 2017 Mouse V1 (2-Photon Ca^{2+}) Comparative Set (Visuomotor Mismatch) *Cell* 169, 1291–1302 2 Bakhtiari et al., 2021 In Silico (Deep Neural Nets) Comparative Set (Hierarchical Streams) *Nat. Commun.* 12, 25164 3 Bastos et al., 2012 In Silico / Theory (Microcircuit) Comparative Set (Canonical Microcircuit) *Neuron* 74, 695–708 4 Bastos et al., 2020 Primate V1/V4/PFC (Laminar ECoG) Comparative Set (Laminar Rhythms) *Neuron* 108, 124–137 5 Bekinschtein et al., 2009 Human (Scalp EEG) Comparative Set (LO vs GO Originator) *PNAS* 106, 1672–1677 6 Chao et al., 2018 Primate Temporal/Frontal (ECoG) Comparative Set (Large-Scale Hierarchy) *Nat. Commun.* 9, 3188 7 Friston et al., 2010 Theory (Free Energy Principle) Canonical Extension (Bayesian Inference) *Nat. Rev. Neurosci.* 11, 127 8 Furutachi et al., 2024 Mouse V1 (2-Photon / Opto) Comparative Set (VIP/SST Disinhibition) *Nature* 633, 398–406 9 Garrett et al., 2020 Mouse V1 (2-Photon Ca^{2+}) Comparative Set (Novelty vs Deviance) *eLife* 9, e50340 10 Greedy et al., 2022 In Silico (Predictive Coding SNN) Comparative Set (Error Backpropagation) *NeurIPS* 35, 24213–24225 11 Hertag et al., 2020 In Silico (Spiking Interneurons) Comparative Set (SST/PV Dynamics) *eLife* 9, e57541 12 Jiang et al., 2024 In Silico (Energy-Based Model) Comparative Set (Sequence Prediction) *PLOS Comput. Biol.* 20, e1011801 13 Keller et al., 2012 Mouse V1 (2-Photon Ca^{2+}) Comparative Set (Sensorimotor Mismatch) *Neuron* 74, 809–815 14 Keller et al., 2018 Synthesis (Cortical Circuits) Comparative Set (Circuit Review) *Neuron* 100, 424–435 15 Kiebel et al., 2008 In Silico (Hierarchical Dynamics) Comparative Set (Temporal Hierarchy) *PLoS Comput. Biol.* 4, e1000209 16 Lao et al., 2023 Human (Auditory Scalp EEG) Comparative Set (Omission Response) *Sci. Adv.* 9, eabq8657 17 Lee et al., 2025 In Silico (Multi-Area Network) Comparative Set (Primate Oscillations) *PLOS Comput. Biol.* 21, e1013469 18 Mikulasch et al., 2023 In Silico / Theory (Plasticity) Comparative Set (Dendritic Error Review) *Trends Neurosci.* 46, 45–59 19 Nejad et al., 2025 Primate / In Silico (Laminar Model) Comparative Set (Layered Routing) *J. Neurosci.* 45, 1234 20 Payeur et al., 2021 In Silico (Dendritic Plasticity) Comparative Set (Burst Multiplexing) *Nat. Neurosci.* 24, 1010–1019 21 Rao et al., 2024 Synthesis (Sensorimotor Neocortex) Canonical Extension (25-Year Synthesis) *Nat. Neurosci.* 27, 1221–1235 22 Rao & Ballard, 1999 In Silico (Hierarchical Model) Canonical Extension (Foundational Model) *Nat. Neurosci.* 2, 79–87 23 Sacramento et al., 2018 In Silico (Dendritic Network) Comparative Set (Somatodendritic Model) *NeurIPS* 31, 8721–8731 24 Spratling et al., 2008 In Silico (Biased Competition) Comparative Set (PC-BC Synthesis) *Biol. Cybern.* 98, 283–292 25 Spratling et al., 2010 In Silico (V1 Microcircuit) Comparative Set (V1 Microcircuit Model) *J. Neurosci.* 30,~~

~~3531–3543 26 Srinivasan et al., 1982 Fly Retina (Intracellular) Canonical Extension (Redundancy Baseline) *Proc. R. Soc. Lond. B* 216, 427 27 VanDerveer et al., 2021 Human (Clinical EEG Cohort) Comparative Set (Sensory Gating Deficits) *Schizophr. Bull.* 47, 1385 28 Wacongne et al., 2011 Human (Auditory MEG) Comparative Set (Hierarchical Rule MMN) *PLoS Biol.* 9, e1001189 29 Wacongne et al., 2012 Human EEG & Spiking Network Comparative Set (Auditory Oddball Model) *J. Neurosci.* 32, 3665–3678 30 Westerberg & Xiong, 2025 Primate V1/V4/IT (Laminar MUA) Primary Index Study (MaDeLaNe) *bioRxiv* 10.1101/2024.10.02 31 Yamins et al., 2014 In Silico / Primate IT (HCNN) Comparative Set (Feedforward Baseline) *PNAS* 111, 8619–8624~~

#	Study Identifier	Modality / Preparation	Inclusion Basis	Bibliographic Citation
1	Attinger et al., 2017	Mouse V1 (2-Photon Ca^{2+})	Comparative Set (Visuomotor Mismatch)	*Cell* 169, 1291–1302
2	Bakhtiari et al., 2021	In Silico (Deep Neural Nets)	Comparative Set (Hierarchical Streams)	*Nat. Commun.* 12, 25164
3	Bastos et al., 2012	In Silico / Theory (Microcircuit)	Comparative Set (Canonical Microcircuit)	*Neuron* 74, 695–708
4	Bastos et al., 2020	Primate V1/V4/PFC (Laminar ECoG)	Comparative Set (Laminar Rhythms)	*Neuron* 108, 124–137
5	Bekinschtein et al., 2009	Human (Scalp EEG)	Comparative Set (LO vs GO Originator)	*PNAS* 106, 1672–1677
6	Chao et al., 2018	Primate Temporal/Frontal (ECoG)	Comparative Set (Large-Scale Hierarchy)	*Nat. Commun.* 9, 3188
7	Friston et al., 2010	Theory (Free Energy Principle)	Canonical Extension (Bayesian Inference)	*Nat. Rev. Neurosci.* 11, 127
8	Furutachi et al., 2024	Mouse V1 (2-Photon / Opto)	Comparative Set (VIP/SST Disinhibition)	*Nature* 633, 398–406
9	Garrett et al., 2020	Mouse V1 (2-Photon Ca^{2+})	Comparative Set (Novelty vs Deviance)	*eLife* 9, e50340
10	Greedy et al., 2022	In Silico (Predictive Coding SNN)	Comparative Set (Error Backpropagation)	*NeurIPS* 35, 24213–24225
11	Hertag et al., 2020	In Silico (Spiking Interneurons)	Comparative Set (SST/PV Dynamics)	*eLife* 9, e57541
12	Jiang et al., 2024	In Silico (Energy-Based Model)	Comparative Set (Sequence Prediction)	*PLOS Comput. Biol.* 20, e1011801
13	Keller et al., 2012	Mouse V1 (2-Photon Ca^{2+})	Comparative Set (Sensorimotor Mismatch)	*Neuron* 74, 809–815
14	Keller et al., 2018	Synthesis (Cortical Circuits)	Comparative Set (Circuit Review)	*Neuron* 100, 424–435
15	Kiebel et al., 2008	In Silico (Hierarchical Dynamics)	Comparative Set (Temporal Hierarchy)	*PLoS Comput. Biol.* 4, e1000209
16	Lao et al., 2023	Human (Auditory Scalp EEG)	Comparative Set (Omission Response)	*Sci. Adv.* 9, eabq8657
17	Lee et al., 2025	In Silico (Multi-Area Network)	Comparative Set (Primate Oscillations)	*PLOS Comput. Biol.* 21, e1013469
18	Mikulasch et al., 2023	In Silico / Theory (Plasticity)	Comparative Set (Dendritic Error Review)	*Trends Neurosci.* 46, 45–59
19	Nejad et al., 2025	Primate / In Silico (Laminar Model)	Comparative Set (Layered Routing)	*J. Neurosci.* 45, 1234
20	Payeur et al., 2021	In Silico (Dendritic Plasticity)	Comparative Set (Burst Multiplexing)	*Nat. Neurosci.* 24, 1010–1019
21	Rao et al., 2024	Synthesis (Sensorimotor Neocortex)	Canonical Extension (25-Year Synthesis)	*Nat. Neurosci.* 27, 1221–1235
22	Rao & Ballard, 1999	In Silico (Hierarchical Model)	Canonical Extension (Foundational Model)	*Nat. Neurosci.* 2, 79–87
23	Sacramento et al., 2018	In Silico (Dendritic Network)	Comparative Set (Somatodendritic Model)	*NeurIPS* 31, 8721–8731
24	Spratling et al., 2008	In Silico (Biased Competition)	Comparative Set (PC-BC Synthesis)	*Biol. Cybern.* 98, 283–292
25	Spratling et al., 2010	In Silico (V1 Microcircuit)	Comparative Set (V1 Microcircuit Model)	*J. Neurosci.* 30, 3531–3543
26	Srinivasan et al., 1982	Fly Retina (Intracellular)	Canonical Extension (Redundancy Baseline)	*Proc. R. Soc. Lond. B* 216, 427
27	VanDerveer et al., 2021	Human (Clinical EEG Cohort)	Comparative Set (Sensory Gating Deficits)	*Schizophr. Bull.* 47, 1385
28	Wacongne et al., 2011	Human (Auditory MEG)	Comparative Set (Hierarchical Rule MMN)	*PLoS Biol.* 9, e1001189
29	Wacongne et al., 2012	Human EEG & Spiking Network	Comparative Set (Auditory Oddball Model)	*J. Neurosci.* 32, 3665–3678
30	Westerberg & Xiong, 2025	Primate V1/V4/IT (Laminar MUA)	Primary Index Study (MaDeLaNe)	*bioRxiv* 10.1101/2024.10.02
31	Yamins et al., 2014	In Silico / Primate IT (HCNN)	Comparative Set (Feedforward Baseline)	*PNAS* 111, 8619–8624

Supplementary Table S5: 31-Paper Literature Corpus Registry with modality, inclusion basis, and provenance.

~~**Model Name Parameters Architecture & Provenance** gemma-3-27b-it 27B~~

~~Google DeepMind (Dense Transformer) gemma-4-31b-it 31B Google DeepMind (Extended Reasoning Architecture) deepseek-r1-distill-llama-70b 70B DeepSeek-AI / Meta Llama Distillation gpt-oss-claude-4.5-sonnet 20B Community Distillation / Reasoning mistral-nemo-12b-thinking 12B Mistral AI & NVIDIA olmo-3-32b-think 32B Allen Institute for AI (Ai2) phi-4-reasoning-plus 14B Microsoft Research qwen3-14b-gemini-3-pro 14B Alibaba Cloud / Community Fine-tune qwen3-14b-claude-4.5-sonnet 14B Alibaba Cloud / Community Fine-tune qwen3.5-40b-claude-4.5-opus 40B Alibaba Cloud / High-Capacity MoE~~

Model Name	Parameters	Architecture & Provenance
gemma-3-27b-it	27B	Google DeepMind (Dense Transformer)
gemma-4-31b-it	31B	Google DeepMind (Extended Reasoning Architecture)
deepseek-r1-distill-llama-70b	70B	DeepSeek-AI / Meta Llama Distillation
gpt-oss-claude-4.5-sonnet	20B	Community Distillation / Reasoning
mistral-nemo-12b-thinking	12B	Mistral AI & NVIDIA
olmo-3-32b-think	32B	Allen Institute for AI (Ai2)
phi-4-reasoning-plus	14B	Microsoft Research
qwen3-14b-gemini-3-pro	14B	Alibaba Cloud / Community Fine-tune
qwen3-14b-claude-4.5-sonnet	14B	Alibaba Cloud / Community Fine-tune
qwen3.5-40b-claude-4.5-opus	40B	Alibaba Cloud / High-Capacity MoE

Supplementary Table S6: Open-weight LLM evaluator models evaluated in this work.

~~**Evaluator Comparison Paired N Pearson r MAE Directional Concordance** Human–Human (H_1 vs H_2) 299 0.390 0.308 88.96% Human Consensus vs LLM Council 175 0.162 0.301 93.14% Council Inter-Model Average 584 0.540 0.284 85.00% Shuffled Evidence (Null H_0) 191 -0.087 0.662 56.54%~~

Evaluator Comparison	Level	Paired N	Pearson r	p	MAE	Concordance
Human–Human (H_1 vs H_2)	Factor cell	299	0.390	< 0.0001	0.308	88.96%
Human Consensus vs LLM Council	Study level	25	0.529	0.0125	—	—
Human Consensus vs LLM Council	Factor cell	175	0.162	0.0322	0.301	93.14%
Council Inter-Model Average	Factor cell	584	0.540	< 0.0001	0.284	85.00%
Shuffled Evidence (Null H_0)	Factor cell	191	-0.087	0.2314	0.662	56.54%

Supplementary Table S7: Evaluator Strata Agreement and Error Hierarchy Summary ($K = 2$ independent human domain experts). The independent study-level comparison ($N = 25$ papers with jointly evaluated factor ratings; Pearson $r = 0.529$, $p = 0.0125$, 95% CI [0.169, 0.764]) treats the independent paper as the unit of analysis. Factor-cell comparisons are descriptive summaries across scored cells nested within papers. Directional concordance denotes agreement in sign on non-zero factor cells.

~~**Adjudication Class Cells (N) Human Supported Council Supported Primary Failure / Divergence Mode** Theoretical Model / Synthesis 15 100.0% (15/15) 86.7% (13/15) Council magnitude attenuation on broad claims Direct Empirical Electrophysiology 7 100.0% (7/7) 71.4% (5/7) Subtle cell-type specificity (SST vs PV gain) Hierarchical / Multi-Area Studies 5 80.0% (4/5) 60.0% (3/5) Conflation of local vs global deviance scope **Total Adjudicated Disagreements 27 96.3% (26/27) 77.8% (21/27) Passage-grounded technical concordance**~~

Adjudication Category	Cells (N)	Proportion	Primary Disagreement / Failure Mode
Directly supported	6	20.0% (6/30)	Full primary text, equation, or figure corroboration of Council assignment
Partially supported	18	60.0% (18/30)	Relevant primary evidence present; conservative magnitude attenuation or specificity nuance
Unsupported attribution	6	20.0% (6/30)	Subtype gain conflation ($N = 3$), laminar error inversion ($N = 1$), untested modality penalty ($N = 1$), cross-paper attribution ($N = 1$)
Total Audited Disagreements	**30**	**100.0% (30/30)**	**Diagnostic audit of** $\lvert H - C\rvert \geq 0.50$ **cells (**$N = 296$ **eligible paired cells)**

Supplementary Table S8: Diagnostic ~~Passage-Support Adjudication across 27 Prespecified Disagreement Cells .~~ Primary-Source Audit across 30 Numerically Selected Large Human–Council Disagreement Cells ($\lvert H - C\rvert \geq 0.50$ among 296 eligible paired cells; 10.1% of paired universe). Evaluates whether large disagreements reflect unsupported evidence attribution or differences in magnitude, contextual interpretation, or biological specificity despite relevant primary-source evidence. Because this diagnostic sample was enriched for large discrepancies, these proportions characterize disagreement modes and do not represent corpus-wide accuracy or hallucination rates.

Decoding Setting / Comparison	Evaluated Units (N)	Correlation	MAE	Paper Ranking Stability
Temperature Stability (gemma-4-31b-it)				
$T = 0.00$ vs. $T = 0.35$	679 factor cells	Pearson $r = 0.9900$	0.0266	Spearman $\rho_{\text{rank}} = 0.9596$
$T = 0.00$ vs. $T = 0.70$	679 factor cells	Pearson $r = 0.9760$	0.0421	Spearman $\rho_{\text{rank}} = 0.9212$
$T = 0.35$ vs. $T = 0.70$	679 factor cells	Pearson $r = 0.9816$	0.0367	Spearman $\rho_{\text{rank}} = 0.9482$
Replicate Reliability (gemma-4-31b-it)				
Replicates at $T = 0.00$	639 triples	Mean $\bar{r} = 0.9905$	0.0126	Spearman $\rho_{\text{rank}} \geq 0.9818$
Replicates at $T = 0.35$	619 triples	Mean $\bar{r} = 0.9724$	0.0384	Spearman $\rho_{\text{rank}} \geq 0.9782$
Replicates at $T = 0.70$	572 triples	Mean $\bar{r} = 0.9444$	0.0585	Spearman $\rho_{\text{rank}} \geq 0.9745$

~~**Experimental Condition**~~**Supplementary Table S9:** ~~**Calls (N) Mean Score ±SD Displacement $\Delta_{\text{LO-GO}}$ Paper Ranking Stability** $T = 0.00$ (Deterministic)~~ Factorial Robustness Analysis across Sampling Temperatures and Stochastic Replicates for the Complete Evaluation Model (`gemma-4-31b-it`, 279~~0.463 ± 0.381 +0.041 Spearman $\rho_{\text{rank}} = 1.000$ (Baseline) $T = 0.35$ (Moderate)~~ /279 ~~0.458 ± 0.384 +0.039 Spearman $\rho_{\text{rank}} = 0.9841$ $T = 0.70$ (Exploratory)279 0.451 ± 0.392 +0.036 Spearman $\rho_{\text{rank}} = 0.9677$ Repeat 1 279 0.460 ± 0.382 +0.040 Spearman $\rho_{\text{rank}} = 0.9818$ Repeat 2 279 0.457 ± 0.386 +0.038 Spearman $\rho_{\text{rank}} = 0.9782$ Repeat~~ factorial conditions). Quantitative factorial statistics are reported exclusively for `gemma-4-31b-it` because it completed full factorial coverage across all 31 papers × 3 temperatures × 3 repeats. Extended chain-of-thought models (`phi-4-reasoning-plus` and `olmo-3-32b-think`) completed 130/279 ~~0.455 ± 0.389 +0.038 Spearman $\rho_{\text{rank}} = 0.9745$ Westerberg & Xiong (2025) — Divergent Outlier $\Delta = -0.400$ Rank 28–29/31 across all T conditions Leave-One-Model-Out (LOMO)10 folds $\Delta_{\text{Total}} \in [-0.105, -0.093]$ Negative throughout H2 $\Delta \in [-0.228, -0.208]$~~

~~**Supplementary Table S9:** Factorial Robustness Analysis across 837 LLM inference calls including Paper Ranking and LOMO Sensitivity~~and 30/279 calls, respectively, but frequently exhausted dynamic generation token budgets during chain-of-thought reasoning prior to emitting the terminal JSON scoring block, precluding equivalent comparative factorial statistics. Paper ranking stability measures Spearman correlation between paper mean scores across conditions ($N = 29$ evaluable papers).

~~**Evidence Manipulation Calls (N) Mean LO Score Mean GO Score Displacement $\Delta_{\text{LO-GO}}$** Intact Evidence (FULL) 18 0.512 ± 0.081 0.474 ± 0.076 +0.038 (LO > GO) Shuffled Evidence (H_0) 18 0.489 ± 0.092 0.493 ± 0.088 −0.004 (Neutral) Prior Only (Zero Evidence) 18 0.450 ± 0.110 0.450 ± 0.108 +0.000 (Null)~~

Evidence Manipulation	**Calls (N)**	**Mean LO Score [±SD]**	**Mean GO Score [±SD]**
Intact Evidence (FULL)	18	0.512 ± 0.081	0.474 ± 0.076
Shuffled Evidence (H_0)	18	0.489 ± 0.092	0.493 ± 0.088
Prior Only (Zero Evidence)	18	0.450 ± 0.110	0.450 ± 0.108

Supplementary Table S10: Evidence Grounding vs. Prior Bias Sensitivity Analysis (54 Calls ~~).~~ across Intact, Shuffled, and Prior-Only Controls). Under intact text, models distinguish Local vs. Global Oddball contexts, whereas under shuffled text (H_0) and prior-only prompts without manuscript text, scores across contexts collapse to parity.

~~**Ablation Condition Paired Cells (N) Pearson r MAE Displacement $\Delta_{\text{LO-GO}}$** FULL Baseline (With Figures) 533 1.000 0.000 +0.0408 NO_FIGURE (Figures Stripped) 533 0.8539 0.2253 +0.0565~~

Ablation Condition	**Paired Cells (N)**	**Pearson r**	**MAE**	**Directional Concordance**
FULL Baseline (With Figures)	533	1.000	0.000	100.0% (Baseline)
NO_FIGURE (Figures Stripped)	533	0.8539	0.2253	99.42% (518/521 non-zero cells)

Supplementary Table S11: Figure-Text-Removal Sensitivity Summary ($N = 31$ papers).

~~**Dimension Stratum Papers (N) Council $\Delta_{\text{LO-GO}}$ Directional Trend** Study Type Empirical 13 −0.009 GO > LO Study Type Computational / Theory 18 −0.013 GO > LO Modality In Silico 19 −0.013 GO > LO Modality Human In Vivo 7 −0.059 GO > LO Modality Animal In Vivo 5 +0.087 LO > GO Publication Era 1982–2014 12 −0.024 GO > LO Publication Era 2015–2020 7 −0.041 GO > LO Publication Era 2021–2025 12 +0.038 LO > GO Publication Status Peer-Reviewed Journal 30 −0.010 GO > LO Publication Status Preprint (Westerberg & Xiong 2025) 1 −0.400 GO ≫ LO~~

Dimension	Stratum	Papers (N)	Council LO Mean	Council GO Mean
Study Type	Empirical	13	0.419	0.428
Study Type	Computational / Theory	18	0.351	0.365
Modality	In Silico	19	0.363	0.376
Modality	Human In Vivo	7	0.357	0.416
Modality	Animal In Vivo	5	0.511	0.424
Publication Era	1982–2014	12	0.368	0.393
Publication Era	2015–2020	7	0.447	0.488
Publication Era	2021–2025	12	0.347	0.309
Publication Status	Peer-Reviewed Journal	30	0.381	0.392
Publication Status	Preprint (Westerberg & Xiong 2025)	1	0.453	0.053

Supplementary Table S12: Corpus Subgroup Sensitivity Analysis across Taxonomic Strata. Demonstrates that council-estimated evidence support across Local and Global Oddball contexts varies by experimental modality and publication era.

~~**Validation Axis Evaluated Figures (N) Correct Partial / Minor Error Overall Accuracy Rate** Axis A: Numerical Trend & Data Extraction 10 9 1 (Minor plot type phrasing) 100.0% PASS (10/10) Axis B: Empirical Effect Direction 10 10 0 100.0% Correct (10/10) Axis C: HPC-36 Ontology Factor Mapping 10 9 1 (Partial invertebrate mapping) 100.0% PASS (10/10) **Overall Manual Validation Outcome 10 Figures 10/10 PASS 0 Fatal Inversions 100.0% PASS**~~

Validation Axis	Evaluated (N)	Strictly Correct	Partial / Minor	Audit Pass Rate
Axis A: Numerical Trend & Data Extraction	10	9	1 (Minor phrasing)	100.0% PASS (10/10)
Axis B: Empirical Effect Direction	10	10	0	100.0% PASS (10/10)
Axis C: HPC-36 Ontology Factor Mapping	10	9	1 (Invertebrate)	100.0% PASS (10/10)
Overall Manual Validation Outcome	**10 Figures**	**10/10**	**0 Fatal Inversions**	**100.0% PASS**

Supplementary Table S13: Manual Expert Validation Summary of Generated Multimodal Figure Descriptions ($N = 10$ ~~prespecified~~ representative figures audited against ~~primary-source PDFs)~~primary source PDFs). All 10 figures met the prespecified audit criterion (Audit Pass Rate: 100.0%); minor phrasing and partial ontology mappings are reported separately.

Context	Hyp.	N	Human Mean	Council Mean	Human−Council [95% CI]	$t(df), p$
Local Oddball	H_1	21	$+0.537 \pm 0.041$	$+0.337 \pm 0.027$	$+0.200$ $[0.111, 0.289]$	$t(20) = 4.69, p = 0.0001$
(LO)	H_2	23	$+0.552 \pm 0.045$	$+0.363 \pm 0.026$	$+0.189$ $[0.117, 0.261]$	$t(22) = 5.42, p < 0.0001$
	H_3	14	$+0.520 \pm 0.034$	$+0.268 \pm 0.045$	$+0.252$ $[0.161, 0.344]$	$t(13) = 5.95, p < 0.0001$
Global Oddball	H_1	18	$+0.600 \pm 0.048$	$+0.299 \pm 0.031$	$+0.302$ $[0.182, 0.422]$	$t(17) = 5.32, p < 0.0001$
(GO)	H_2	22	$+0.560 \pm 0.043$	$+0.259 \pm 0.049$	$+0.300$ $[0.205, 0.396]$	$t(21) = 6.54, p < 0.0001$
	H_3	11	$+0.557 \pm 0.065$	$+0.312 \pm 0.057$	$+0.244$ $[0.080, 0.408]$	$t(10) = 3.32, p = 0.0077$

Supplementary Table S14: Primary Paper-Level Hypothesis Calibration and Concordance ($K = 2$ Human Consensus vs. Autonomous Council). Analysis is restricted to independent papers where both human experts ($K = 2$) evaluated mechanisms within the specific hypothesis. Council models were averaged within paper first. SEM denotes standard error across independent papers ($\mathrm{SD}/\sqrt{N}$). Paired differences (Human−Council mean difference) evaluate systematic score attenuation. Paired t-tests ($t(df)$) and two-sided p-values evaluate the difference in mean support.

Omitted Evaluator Model	Paired N	Pearson r	p	Spearman ρ	p	MAE
`deepseek-r1-distill-llama-70b`	1710	0.9553	< 0.0001	0.9589	< 0.0001	0.0465
`gemma-3-27b`	1733	0.9747	< 0.0001	0.9472	< 0.0001	0.0408
`gpt-oss-claude-4.5-sonnet`	1801	0.9920	< 0.0001	0.9844	< 0.0001	0.0086
`mistral-nemo-12b-thinking`	1775	0.9732	< 0.0001	0.9698	< 0.0001	0.0239
`olmo-3-32b-think`	1798	0.9900	< 0.0001	0.9800	< 0.0001	0.0122
`phi-4-reasoning-plus`	1778	0.9825	< 0.0001	0.9735	< 0.0001	0.0255
`qwen3-14b-gemini-3-pro`	1800	0.9797	< 0.0001	0.9649	< 0.0001	0.0243
`qwen3.5-40b-claude-4.5-opus`	1724	0.9497	< 0.0001	0.9178	< 0.0001	0.0440
Evaluator profiles with hypothesis-level scores but unassigned individual factor cells:						
`gemma-4-31b`	1816	—	—	—	—	—
`gpt-oss-safeguard-120b`	1816	—	—	—	—	—

Supplementary Table S15: Leave-One-Model-Out Council Score Stability Analysis across Evaluator Architectures. Evaluates the sensitivity of aggregate Council scores to single-model removal by comparing the complete Council continuous mean against the Council recomputed after omission of each contributing evaluator across matched factor cells ($N = 1710$–1801). Across all active evaluator omissions ($N = 8$ model omissions), Council scores remained highly stable: mean Pearson $r = 0.9747$ (95% CI: $[0.9619, 0.9874]$, range 0.9497–0.9920), mean Spearman $\rho = 0.9621$ (95% CI: [0.9441, 0.9800], range 0.9178–0.9844), and mean MAE $= 0.0282$ (95% CI: [0.0163, 0.0402], range 0.0086–0.0465). Confidence intervals are computed using Student's t-distribution ($df = N_{\text{omissions}} - 1 = 7$) with model omission as the unit. MAE is reported in original $[-1.0, +1.0]$ score units. Two evaluator profiles contributed hypothesis-level Council scores but no individual factor-cell scores and therefore were not evaluable in this factor-level leave-one-out analysis.